# SILAS KWABLA GAH

**PhD Candidate in Computer Science · AI Control, Multi-Agent Oversight & Verifiable Reinforcement Learning**

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## Research Summary & Scientific Background

Computer Scientist and AI Systems Researcher specializing in **AI control**, **verifiable reinforcement learning (RLVR)**, **multi-agent oversight mechanisms**, and **probabilistic calibration for heterogeneous foundation models**. My doctoral and applied research focuses on designing oversight signals that remain robust, ungameable, and informative when capable models or multi-agent teams optimize against them. Methodological highlights include:

- **Verifiable Oversight & Anti-Reward-Hacking**: Developed label-free process rewards from frozen multi-view perception models with a *two-arm validate-before-train protocol* that uncovered specification gaming (area inflation) and rectified it via count-based precision and weighted $F_{0.5}$ metrics in online **GRPO/VERL** on **Qwen2.5-VL-7B**.
- **Calibrated Foundation Model Arbitration**: Formulated **RCASA** (*under review at IEEE TPAMI*), pairing Platt confidence calibration with Bernoulli uncertainty estimation to dynamically arbitrate heterogeneous model evidence at inference time ($\Delta\theta = 0$).
- **Agentic Systems Orchestration**: Architected multi-actor state-graph systems using **LangGraph** and **LangChain** with real-time asynchronous watchdog evaluation (**DeepEval**) and prompt tracing observability (**Opik/Comet ML**).

## Core Research Contributions

**Label-Free Process Rewards & De-Risked RLVR** *2025 – Present*
*Lead Researcher · Verifiable RL, Anti-Hacking & Multi-Modal Reasoning*

- Formulated a label-free process reward utilizing cross-view geometric agreement among frozen perception backbones (RegionPLC, SAM3), eliminating dependence on expensive human labels or learned reward models.
- Designed a **Two-Arm Validate-Before-Train Protocol**: Arm I verified 100% pairwise discrimination on ScanRefer; Arm II subjected the reward to adversarial counterfactuals, discovering an enumeration failure mode where oversized boxes gamed recall.
- Redesigned the reward around count-based precision and Weighted $F_{0.5}$ ($\beta = 0.5$), eliminating area bias ($\rho > 0.72 \to 0.334$) and demonstrating full system compatibility with online **Group Relative Policy Optimization (GRPO)** in **VERL**.

**Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)** *2025 – Present*
*Lead Author · Submitted to IEEE TPAMI (Major Journal)*

- Developed a probabilistic framework for coordinating multi-source foundation model outputs using **Platt scaling**, **Bernoulli uncertainty estimation**, and **contextual reliability weighting**.
- Established theoretical bounds and empirical oracle headroom metrics; achieved training-free SOTA on open-vocabulary benchmarks (**77.9% HM** on PASCAL-$5^i$, **33.68% HM** on ScanNet200).

**Generalized Few-Shot Open-Vocabulary Arbitration** *2025 – 2026*
*Lead Author · Submitted to WACV 2026 & ACCV 2026*

- Built training-free semantic arbitration across frozen 2D/3D vision-language models (SAM3 + CLIP + DINOv2), outperforming fully fine-tuned baselines by **+27.2%** Novel mIoU.

## Agentic AI Systems & Multi-Actor Workflows

**Ghana Legal AI — Production Multi-Agent RAG Platform** *2024 – Present*
*Lead Architect · LangGraph, LangChain, MongoDB Atlas Vector Search, Groq Llama 3*

- Engineered a state-graph multi-agent workflow via **LangGraph** coordinating 3 specialized personas with stateful checkpointing.
- Implemented real-time watchdog evaluation using **DeepEval** (Faithfulness, Relevancy, Hallucination) and full telemetry via **Opik**.

**LLM Twin — End-to-End LLM Lifecycle & Optimization** *2024 – 2025*
*ML Engineer · ZenML, AWS SageMaker, Qdrant, Llama 3.1*

- Implemented complete LLM lifecycle: ETL data crawlers → feature store → **SFT and DPO fine-tuning of Llama 3.1** → Qdrant vector retrieval.

## Education

**PhD Candidate, Computer Science** *2020 – Dec 2026 (Expected)*
*University of Ghana, Legon, Accra, Ghana*

- *Thesis:* Reliability-Calibrated Adaptive Semantic Arbitration for Inference-Time Adaptation of Frozen Foundation Models
- *Supervisor:* Prof. Ebenezer Owusu

**MPhil, Computer Science** *2016*
*University of Ghana, Legon, Accra, Ghana*

- *Thesis:* Sentiment Analysis of Twitter Feeds using Machine Learning: Effect of Feature Hash-Bit Size

**BSc, Computer Science** *2012*
*GIMPA, Accra, Ghana*

**HND, Electrical Engineering** *2009*
*Accra Technical University, Accra, Ghana*

## Publications & Manuscripts

1. **Gah, S. K.** et al. (2026). Cross-View Agreement as a Label-Free Process Reward for 3D-Grounded Reinforcement Learning. *Under Submission to SACAIR 2026* [Targeting Multimodal RLVR].
2. **Gah, S. K.**, Owusu, E. et al. (2026). When Does Inference-Time Semantic Arbitration Help? Reliability, Complementarity, and Oracle Headroom across 2D and 3D Segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)* [Under Review].
3. **Gah, S. K.** et al. (2026). Training-Free Generalized Few-Shot Segmentation through Open-Vocabulary Semantic Arbitration. *WACV 2026* [Under Review].
4. **Gah, S. K.** et al. (2026). Training-Free Open-Vocabulary 3D Point-Cloud Segmentation on the Generalized Few-Shot Benchmark. *ACCV 2026* [Under Review].
5. Appati, J. K., Boante, L. M., Owusu, E., & **Gah, S. K.** (2023). Transfer Learning-Based Encoder-Decoder Model for Skin Lesion Segmentation. *ICICCT 2023*, CCIS Vol. 1841, Springer.

## Technical Skills

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| **AI Safety & RL** | RLVR, GRPO, PPO, DPO, Process Rewards, Reward Hacking Mitigation, Counterfactual Testing, VERL. |
| **Foundation Models** | Qwen2.5-VL, Llama 3.1, SAM3, RegionPLC, DINOv2, CLIP, SFT, LoRA/QLoRA, vLLM, OpenRouter, DeepEval, Opik. |
| **Agentic AI & Systems** | LangGraph, LangChain, Multi-Agent Orchestration, Qdrant, MongoDB Atlas Vector Search, ZenML. |
| **Languages & Cloud** | Python (Expert), PyTorch, CUDA, C++, Java, TypeScript, SQL, Docker, Kubernetes, AWS SageMaker, GCP, Git. |

## Academic References

1. **Prof. Ebenezer Owusu** — PhD Supervisor, Department of Computer Science, University of Ghana (ebeowusu@ug.edu.gh)
2. **Dr. Emmanuel Owusu-Oware** — Senior Lecturer, Department of IT Studies, UPSA, Accra, Ghana (emmanuel.owusu-oware@upsamail.edu.gh)
3. **Dr. Rose Atuah** — Assistant Professor, Colorado State University–Pueblo, USA (rose.atuah@csupueblo.edu)